Actual geometric torsion for the BK test curve

For $E/F5: y^2 = x^3 + 1$ and $r = 3$, Lean proves that every geometric3-torsion point descends from the actual field F25. Over every algebraically closed characteristic5 extension K, E(K)[3] has exactly nine points and an actual additive basis indexed by $(Z/3)^2$. Frobenius acts as $(a,b) \mapsto (a,-b)$, and the actual F5-rational image is exactly $b=0$. [pcc:unverified]

The finite basis is proved before transport. A general torsion-rank theorem, an assumed basis, and a source nondegeneracy axiom are not premises. The earlier issued Miller packet and its historical scope are unchanged. [pcc:unverified]

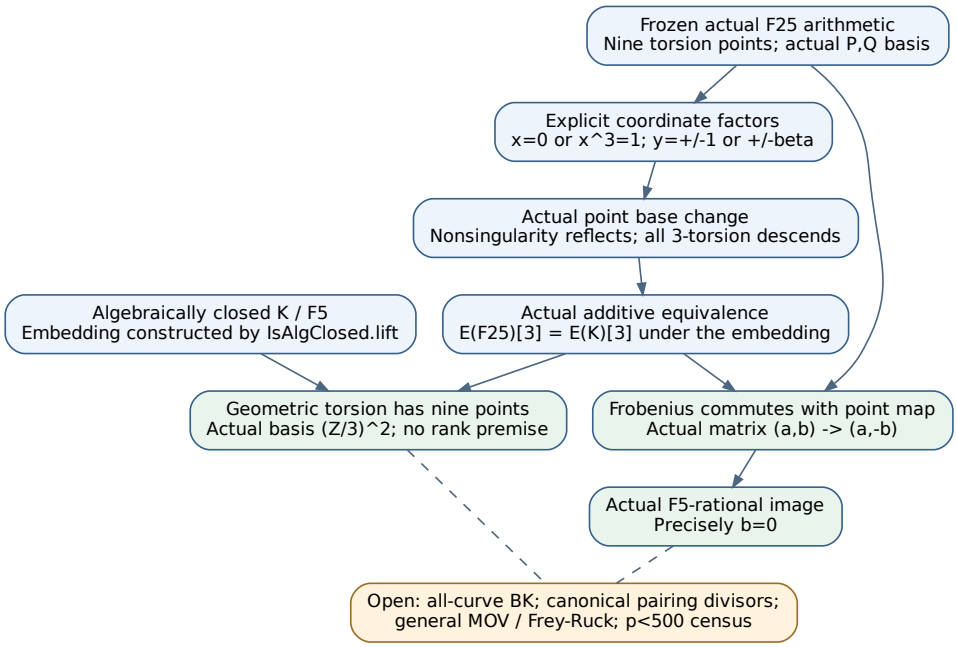

Figure 1: Coordinate descent, actual geometric basis and scope. [pcc:unverified]

Proof map, hypotheses and assurance

Let $\beta^2=2$ and $\omega=2+3\beta$ in the actual F25. The frozen actual3-torsion classification gives $x=0$ or $x^3=1$. Kernel-checked finite identities give $\omega^3=1$ and $\omega^2+\omega+1=0$. Over any F5-algebra embedding $f:F25\to K$, the universal identity $(x-1)(x-f(\omega))(x-f(\omega)^2)=x^3-1$ therefore forces x to descend. The actual curve equation forces $y=\pm1$ when $x=0$, and $y=\pm f(\beta)$ otherwise. Both coordinates descend. [pcc:unverified]

Nonsingularity reflects through f. The actual point map is injective, so $3R=0$ reflects to the lifted point. Its restriction to actual3-torsion is bijective and additive. The previously checked actual F25 basis transfers through this actual homomorphism, proving cardinality9. For algebraically closed K, Mathlib constructs f with IsAlgClosed.lift from the finite algebraic extension; no embedding hypothesis remains. [pcc:unverified]

Frobenius commutes with the point map because f preserves powers. Transporting the checked F25 action gives $(a,b)\mapsto(a,-b)$ on the geometric basis. The actual finite-field descent criterion equates rational image with Frobenius-fixed points, giving precisely $b=0$. [pcc:unverified]

Module	Checked structural obligation
CoordinateDescent	Actual root identities plus universal factorization and both coordinate descents
GeometricTorsion	Actual torsion point-map surjectivity, additive equivalence, cardinality and basis
AlgebraicClosure	Embedding constructed in every algebraically closed characteristic5 extension
GeometricFrobenius	Actual Frobenius commutation, matrix, rational image and combined certificate

The full project proof/axiom gate and exact statement/source pins are recorded in sources.json. All failed source versions and build exits are retained in lossless archives. Lean4.35.0 and Mathlib56a00585 are pinned. Only propext, Classical.choice and Quot.sound are admitted; no holes, custom axioms, native_decide, trusted-native exceptions, waived linters or new elaboration options. Actual local replay and independently signed standing are recorded separately in verification.json. [pcc:unverified]

Sources: frozen actual BK coordinate and finite Miller arithmetic proofs; prior selected Galbraith chapter26 domain reading; pinned Mathlib actual point map/injectivity, nonsingularity reflection and IsAlgClosed.lift. KB unavailable (MCP absent/wrapper127), bounded repository/primary/Mathlib fallback. No novelty or new source-refutation claim. [pcc:unverified]

This is one concrete curve and $r=3$. Generic all-curve BK rank/determinant, the $p<500$ census, canonical principal-divisor/valuation pairing identification and generic MOV/Frey–Rück remain open. No native experiment, benchmark, timing, independent-host or paid-compute claim. [pcc:unverified]